

Chapter 3 - Harish-Chandra Induction

Recall/Learn: The forgetful functor

$$\text{AbGrp} \rightarrow \text{ComMon} = \text{commutative monoids}$$

has a left adjoint

$$G: \text{ComMon} \rightarrow \text{AbGrp}$$

called the group completion, defined as follows

$$G(M) = M \times M / \sim$$

where $(m_1, n_1) \sim (m_2, n_2)$ if $\exists k \in M$ st.

$$m_1 + n_2 + k = m_2 + n_1 + k$$

So equiv class of (m, n) is $m - n$.

$$\text{Eg } G(\mathbb{N}) = \mathbb{Z}$$

Now let R be any ring.

$$R\text{-Mod} := \text{cat of fin gen left } R\text{-mods}$$

Given $P \in R\text{-Mod}$ projective, write $[P]$ for its iso class. Then the set of iso classes $[P]$ of fin gen proj R -mods is a monoid under

$$[P_1] + [P_2] := [P_1 \oplus P_2]$$

The Grothendieck group of R , denoted $K_0(R)$, is the group completion of this monoid.

In fact, if we have a ring map $R \rightarrow S$ and P is a fin gen proj R -mod, then $S \otimes_R P$ is a fin gen proj S -mod, and K_0 is a functor

$$K_0: \text{Ring} \rightarrow \text{AbGrp}$$

Eg (Exercise) If R is a Dedekind domain, then

$$K_0(R) \cong \mathbb{Z} \oplus \text{Cl}(R)$$

Main Eg Let Γ be a fin group, $\mathbb{C}\Gamma = \text{group alg over } \mathbb{C}$ of Γ . Then

$$K_0(\mathbb{C}\Gamma) \cong \text{free ab group on irreps of } \Gamma$$

$$\cong \text{free ab group on irred characters}$$

so $K_0(\mathbb{C}\Gamma) \otimes \mathbb{C} \cong \text{vect space of class functions}$

$$\phi: \Gamma \rightarrow \mathbb{C}$$

Indeed any fin dim Γ -rep on a $\mathbb{C}$ -vect sp is a direct sum of irreducibles, and any irreducible rep is a direct summand of $\mathbb{C}\Gamma$

So every fin gen $\mathbb{C}\Gamma$ -mod is proj and

$\{\text{iso classes of fin gen proj } \mathbb{C}\Gamma\text{-mods}\} \cong \text{free commutative monoid on iso classes of irreps}$

$$K_0(\mathbb{C}\Gamma) = \text{free ab group on iso classes of irreps.}$$

On the other hand, we know the irred characters form a basis for the vector space of class functions.

The pairing on $\mathbb{C}\Gamma\text{-Mod}$

$$\langle V_1, V_2 \rangle_\Gamma := \dim_{\mathbb{C}} \text{Hom}_{\mathbb{C}\Gamma}(V_1, V_2)$$

extends to a bilinear pairing $\langle, \rangle_\Gamma: K_0(\mathbb{C}\Gamma) \times K_0(\mathbb{C}\Gamma) \rightarrow \mathbb{Z}$

Under the identification $K_0(\mathbb{C}\Gamma) = \mathbb{Z}$ -lin comb of irred char in the space of class function, this $\langle, \rangle_\Gamma$ is identified with

$$(\phi, \psi) \mapsto \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \phi(g) \psi(g^{-1})$$

For everything above, there was nothing too special about $\mathbb{C}$, and everything works replacing $\mathbb{C}$ with a field K satisfying

Running Assumption $\text{char}(K) = 0$ and K contains all $|\Gamma|$ th roots of 1

Let $K\Gamma = \text{group alg of } \Gamma \text{ over } K$.

Say Γ' is another group. Let M be a $(K\Gamma, K\Gamma')$ -bimodule, finite dimensional over K , i.e. M is a fin dim K -vect space with linear left, resp. right, Γ , resp Γ' , actions.

Then we get a functor

$$F_M: K\Gamma'\text{-Mod} \rightarrow K\Gamma\text{-Mod}$$

$$V \mapsto M \otimes_{K\Gamma'} V$$

If we set $M^* = \text{Hom}_K(M, K)$, is a $(K\Gamma', K\Gamma)$ -bimodule via $f \in M^*, g \in \Gamma, g' \in \Gamma'$, then $(g'f)(m) = f(mg')$ and $(fg)(m) = f(gm)$

Then we also have

$$F_{M^*}: K\Gamma\text{-Mod} \rightarrow K\Gamma'\text{-Mod}$$

$$W \mapsto M^* \otimes_{K\Gamma} W$$

Then (F_M, F_{M^*}) and (F_{M^*}, F_M) are both adjoint pairs:

$$\text{Hom}_{K\Gamma}(F_M(V), W) = \text{Hom}_{K\Gamma}(M \otimes_{K\Gamma'} V, W)$$

$$(\text{Tensor-Hom adjunction})^{\pm 1} = \text{Hom}_{K\Gamma'}(V, \text{Hom}_{K\Gamma}(M, W))$$

$$= \text{Hom}_{K\Gamma'}(V, M^* \otimes_{K\Gamma} W)$$

$$= \text{Hom}_{K\Gamma'}(V, F_{M^*}(W))$$

and

$$\text{Hom}_{K\Gamma'}(F_{M^*}(W), V) = \text{Hom}_{K\Gamma}(W, F_M(V))$$

$$= \text{Hom}_{K\Gamma}(W, F_M(V)) \text{ since } \dim_K M < \infty$$

These functors define homomorphisms

$$F_M: K_0(K\Gamma') \rightarrow K_0(K\Gamma), F_{M^*}: K_0(K\Gamma) \rightarrow K_0(K\Gamma')$$

and then our pairing $\langle, \rangle_\Gamma$ on $|C_0(K\Gamma)$ satisfies

$$\langle F_m(x'), x \rangle_\Gamma = \langle x', F_{m^*}(x) \rangle_{\Gamma'}$$

Eg If Γ' is a subgroup of Γ , and $M = K\Gamma$ as a $(K\Gamma, K\Gamma')$ -bimodule. Then

$$F_m(V) = K\Gamma \otimes_{K\Gamma'} V = \text{Ind}_{\Gamma'}^{\Gamma} V$$

And since $(K\Gamma')^* \cong K\Gamma$ a $(K\Gamma', K\Gamma)$ -bimed

$$f \mapsto \sum_{g \in \Gamma} f(g^{-1})g$$

we have

$$F_{m^*}(W) = K\Gamma \otimes_{K\Gamma'} W = \text{Res}_{\Gamma'}^{\Gamma} W$$

Then $\langle F_m(x'), x \rangle_\Gamma = \langle x', F_{m^*}(x) \rangle_{\Gamma'}$

$$\langle \text{Ind}_{\Gamma'}^{\Gamma} x', x \rangle_\Gamma = \langle x', \text{Res}_{\Gamma'}^{\Gamma}(x) \rangle_{\Gamma'}$$

is Frobenius reciprocity